

Sharpening what is already known

A deep feature-research campaign for the HKJC production model:
handicap state, sectional physics, race shape, and dynamic information quality

Feature Research Phase

31 August 2026

What this document is. A feature-research campaign against an existing production winner-probability model, which it treats as strictly read-only. Four mechanisms were pushed hard: historical handicap / class / rating state, sectional and 200m physical state, field-level pace and race shape, and dynamic latent state with run-specific information quality. The objective was a small number of genuinely strong pre-race features.

What it does not establish. No profitability claim. Every price used is a result-page price recorded after the pool closed, at which no bet could have been placed; no dividends were loaded and no return is computed anywhere. Every season in the archive has been development data, here and in the reference project, so no window is genuinely unseen.

The one-line answer. The campaign's most valuable output is not a feature. It is the discovery that *both* cheap screens it inherited are close to useless at this effect size. The market-residual battery is **invalid** — 83 of 140 candidates pass it, **none** pass the correct test, and 65 change sign. And forward correlation against the next performance ranks candidates roughly in **reverse** of how they behave in production: the direction with the largest forward signal contributed nothing at canonical scale, the one with the smallest contributed the only survivor, and two of six blocks made the model worse.

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1 Executive summary

The production model’s entire edge over the market is -0.004533 of race log loss. That is the yardstick. A feature is worth having only if it moves a meaningful fraction of it.

Four directions were pushed to their own ceilings, five cross-direction crossings were built, and two adversarial checks were added that the brief did not ask for. The results divide unevenly, and the division is the finding.

What is real and useful

- **Pace composition $\times$ own style is the only candidate that survived canonical scale.** -0.001490 of race log loss on 8 years $\times$ 5 seeds $\times$ 400 rounds, $t = -1.25$, improving 25 of 40 paired cells and 6 of 8 years — **33%** of the reference project’s entire edge over the market. It is the most novel finalist against the incumbent (68%) and the only block that retained its development effect (103%). The mechanism is understood: production carries pace pressure and running style separately but not their interaction, and a field-common quantity cancels from the softmax entirely, so the interaction is the *only* form in which race shape can enter the model.
- **The pace is forecastable, and much better than previously thought.** A probabilistic four-class behaviour model plus a full field-composition feature set forecasts the race’s early pace at out-of-sample $r = +0.680$, $R^2 = 0.459$, against 0.032 for the crude prior attempt and 0.414 for Study II’s proper one. The previous failure to convert pace into a useful feature was *not* a forecast-quality problem — and this forecast is what makes the surviving block work.
- **The existing handicap_residual_mean has the largest forward signal in three studies — and it does not survive.** Forward correlation $+0.3763$, positive in 7 of 7 years, $+0.01757$ over a plain recency baseline, about as much as all eleven features this campaign built for Direction 1 combined, and a mechanism read from source. At canonical scale it is -0.000113 with $t = -0.11$ and 4 of 8 years, and the version with its uncertainty companions — which forward evidence ranked highest of anything tested — is **worse than doing nothing** at $+0.001584$. Its novel share against the incumbent is only 29%: `rating_z`, `rating_change`, `class_change` and the price between them rebuild most of it. **An earlier draft of this report recommended it. Canonical confirmation reversed that, which is what canonical confirmation is for.**
- **Where the handicap state and the physical state disagree, the handicap state wins — monotonically.** Across quartiles of disagreement the handicap state’s forward correlation rises $+0.3734 \rightarrow +0.4037$ while the physical state’s falls $+0.3721 \rightarrow +0.2861$. The ordinal, opponent-adjusted construction that Direction 1 set out to improve on physical grounds beats the physical measurement precisely where they conflict.

What is real and too small

200 m sub-split state adds $+0.0018$ over the 400 m bands on identical rows, against a permuted control of -0.0008 — and almost all of it is *peak velocity* ($+0.00308$ of $+0.00314$), not the finishing-shape story that motivated the measurement. Sectional physical state adds $+0.0024$ over recency. Pace composition $\times$ own style adds $+0.0047$.

What failed, and why

Direction 1’s central hypothesis. A residual of physical performance against the official handicap correlates **0.9947** with a plain 60-day decayed mean of performance, because the official handicap explains only **1.4%** of within-race performance variance. There is almost nothing to residualise against.

Direction 2’s central hypothesis. Decomposing sectional performance into style and ability made every one of eight dimension $\times$ half-life combinations *worse*. The reason is measurable: normalising a split against the field’s own pace in that band already removes most of the style, so an explicit style model on top removes disproportionately ability.

Direction 4’s central premise, inverted. With *perfect* identification of compromised runs, downweighting them makes the state **worse**: -0.0012 halved, -0.0119 near-zero, **-0.0282** excluded. A trouble-affected bad run still retains $+0.29$ of itself into the next start. It is partly noise, not only noise.

The reversal

x1_confirmed — the product of the two standardised states — carried **$t = +15.47$** against the market, the largest such statistic in three studies. It is an artifact. The plain square of a *single* state, with no crossing at all, reproduces it at $t = +17.50$; beside the market logit’s own quadratic it becomes **$t = -0.14$** while that quadratic carries $+28.99$; and the effect is symmetric across both tails, which is the convexity signature and not a confirmation signature.

That prompted re-testing every market claim under the production architecture’s own functional form — a racewise softmax with $\log q$ as a fixed offset, which only *within-race* variation can move. Of 140 pre-race candidates: **83 clear $|t| > 3$ under the old battery, 0 under the correct one, and 65 change sign.** **class_vs_best**, reported earlier in this campaign as evidence that the market under-reacts to class droppers ($t = -8.25$), is $t = -2.24$ correctly specified, and is withdrawn.

Conversely the correct test *raises* Direction 3: **prior_style** goes from $t = -0.27$ (i.e. nothing) to **$+2.99$** , the largest legitimate residual in the campaign, worth -0.0012 of race log loss out of sample for a single linear term — 26% of the whole production edge. But nothing clears the pre-registered survivor criterion, and §10 shows those very features are the *most* redundant with the incumbent (**prior_style** novel share 0.203, **own_speed_vs_field** 0.160). Production already contains **early_style_ewm**, **pace_pressure** and **closer_setup**.

The single recommendation

Test pace composition $\times$ own style — canonical block C4 — against the locked production configuration. It is the only candidate in this campaign that survived canonical scale, the most novel of the finalists against the incumbent’s 28 columns, and the only one whose development effect carried over intact. $t = -1.25$ is not significance; it is the best production-scale evidence produced across three studies, so the right next step is a properly powered A/B rather than a promotion.

And the campaign’s transferable lesson is about method, not features. The ranking at canonical scale is close to the *reverse* of the ranking on forward correlation. Direction 1 had the largest forward signal and returned $t = -0.11$; Direction 3 had the smallest and produced the only survivor. Two of the six blocks made the production model *worse*. Combined with §9’s invalidation of the market-residual screen, the honest conclusion is that the two cheap instruments this campaign inherited — forward correlation and the linear-probability market residual — are

both close to useless at this effect size, and only paired canonical confirmation decides anything.

2 Prior research and remaining questions

Two studies preceded this campaign, and their governing findings shaped its design more than the brief did.

Study I (static latent performance) established the geometry: HKJC times races in 400 m sections back from the finish with the first carrying the remainder, valid on 1.0000 of 13,309 races. It established the target: y_z^{rel} , the within-race z -score of log average speed, the most persistent quantity available (self-persistence $r = 0.278$, split-half reliability 0.822). And it established the awkward part: the sectional profile is *not* low-dimensional, and what persists in it is **running style**, not ability.

Study II (dynamic and group state) built crossed random effects by EM, exact quadrature win probabilities, and a continuous-time Kalman filter with heteroskedastic observation noise. Its most useful result was negative: an oracle reliability built from each run’s realised surprise was worth +0.106 of forward correlation, while the best observable model reached **0.27% of it**.

2.1 The two findings that governed this campaign’s method

Development scale carries essentially no information at this effect size. Eight candidates went from a development profile to canonical scale across the two studies. None confirmed. Retentions: 43%, 15%, −23%, 2%, −155%, −401%, −567%, +427%. This is why six blocks went to canonical here rather than sixty.

Novelty and usefulness are inversely related. Spearman +0.39 between novel share and uselessness in Study II. §10 reconfirms it for a third time: the campaign’s most novel construction (`x1_abs_disagree`, novel share 0.837) has forward $r = +0.095$, and its strongest market residual (`own_speed_vs_field`, novel share 0.160) is the most redundant thing in the list.

2.2 What remained open

The reference project’s own research memory (Topic 1) lists Direction 3’s exact feature set as *untested hypotheses*, with the explicit instruction “do not promote these based on intuitive signs” — so Direction 3 was built from that list rather than invented. Topic 2 locates cold-start at 0–3 HK starts (20.0% of rows have ≤ 3), which is why every incremental table here carries a history-depth profile. Topic 6 fixed the reliability and missingness conventions used throughout.

3 Existing feature architecture, and what it means for a candidate

The production architecture is `softmax_offset`. The market’s normalised implied probability enters XGBoost as a **fixed additive base margin** $\log q$, and the booster learns a per-runner correction against a racewise winner-softmax likelihood:

$$u_{ri} = \log q_{ri} + c_{\theta}(x_{ri}), \quad p_{ri} = \text{softmax}_i(u_r).$$

At $c = 0$ the model *is* the market exactly, because q already sums to one within the race. The gradient is $p - y$ and the Hessian the diagonal $p(1 - p)$.

Three consequences govern everything in this report.

A quantity common to the whole field cannot help. The softmax is invariant to adding a constant within a race. This is why the projected pace *level* is worth +0.00015 and the pace $\times$ own-style interaction is worth +0.00465: the level cancels and only the differential response survives. It is also why §9’s corrected market test is the right one — it identifies from within-race variation only, by construction.

The 28 production features cover more ground than a candidate list suggests. `configs/features/pro` includes `early_style_ewm` (a running-*position* habit), `pace_pressure`, `closer_setup`, `trip_excuse_ewm`, `rating_change`, `class_change`, and three barrier-trial columns. Any style, pace-pressure or rating-change candidate must beat those, not beat nothing.

Nothing timed is in production. No sectional time, no split-derived speed, no handicap-state column. That is the gap this campaign probed, and the honest summary of §5 and §12 is that the gap is mostly real and mostly small.

3.1 What `handicap_residual_mean` actually is

Read from source (`src/hkjc/features/horse_state.py:398--436, :724--750, :816--830, :193--199`) rather than inferred, because the name is misleading. It is **not** a residual on physical performance. It is a Kalman-filtered state over a *pairwise finishing-order* residual against the official handicap:

1. $\text{net}_i = R_i - w(W_i - \text{median}_r W)$ — official rating adjusted for weight carried relative to the field median.
2. For each unordered pair, $\hat{p}_{ij} = \sigma((\text{net}_i - \text{net}_j)/s)$ — a Bradley–Terry expectation from the handicapper’s own numbers.
3. Observation $o_{ij} - \hat{p}_{ij}$, aggregated over a horse’s pairs at weight $1/(n - 1)$, with a binomial pair variance accumulated alongside.
4. Filtered through a **six-dimensional** state with context vector $[1, \text{AWT}, \log(d/1400), \log^2(d/1400), \text{HV turf}, \text{so it already carries horse} \times \text{surface}, \text{horse} \times \text{distance (quadratic)}, \text{horse} \times \text{venue and horse} \times \text{class slopes}]$.
5. Half-lives 540 d (intercept), 360 d (context slopes), 75 d (short-run form).

Two properties of this construction explain most of Direction 1’s results. The observation is **ordinal** — a nose and ten lengths are the same observation — which is what motivated building a physical alternative. And it is **opponent-adjusted by design**, which is what §8 shows the physical alternative cannot match.

Why it is not in production. From `docs/research/findings.md`: the horse-memory blocks were tested *as a family*, and “actual-model ablation found no stable incremental value after the market and other feature blocks were present”; “rank state improved winner residual AUC but worsened log loss”; “unrestricted combinations overfit 2025.” It has never been tested alone.

4 Handicap, rating and class state

4.1 The central hypothesis, and its failure

The brief’s premise: the existing state’s observation is ordinal and saturating, so a residual on *physical* performance should be strictly more informative. Define

$$D_{ir} = Y_{ir} - \hat{E}[Y_{ir} \mid R, C, W]$$

with three materially different expectation models, each fitted forward — **linear** (on within-race net-rating difference, weight, class, rating), **withinrace** (on the within-race z-score of net rating plus the field’s rating spread, because *absolute* rating means different things in different fields), and **boosted**.

The result: `D_withinrace_ewma_60` correlates **0.9947** with a plain 60-day decayed mean of performance. The reason is a single number — **the official handicap explains 1.4% of within-race performance variance**. Subtracting an expectation that explains 1.4% of the thing leaves 98.6% of the thing. There is almost nothing to residualise against, and the elaborate construction reduces to recency with extra steps.

This is the campaign’s clearest example of a well-motivated idea whose premise is empirically absent, and it was detectable in one correlation before any of the state machinery was built.

4.2 The handicapper’s lag is real, and does not convert

$\text{corr}(\text{physical change now, rating change now}) = +0.0074$; at the *next* rating change $+0.0726$; at the *previous* one **-0.167**. The handicapper reacts to real performance change late, measurably and in the expected direction. But neither `handicapper_lag` nor its cumulative version converts into more than $+0.008$ of forward correlation, and both are largely reconstructible from `rating_change`, which production already has.

4.3 The verdict

Fourteen nested blocks on a common complete-case sample of 91,904 rows, fitted before 2020 and scored 2020–2026.

Block	k	oos r	Gain	year range
baseline: recency 60 + 180	0	0.34470	—	[0.337, 0.363]
+ <code>handicap_residual_mean</code> (existing)	1	0.36227	+0.01757	[0.350, 0.392]
+ that state and its uncertainty (existing)	3	0.36963	+0.02494	[0.356, 0.407]
+ <code>rank_ability</code> (existing)	1	0.34573	+0.00103	[0.336, 0.367]
+ both existing states	2	0.36556	+0.02086	[0.354, 0.394]
+ all eight existing state columns	8	0.38161	+0.03692	[0.370, 0.416]
+ class history (built here)	4	0.35698	+0.01228	[0.348, 0.365]
+ rating history (built here)	4	0.35551	+0.01081	[0.346, 0.371]
+ handicapper lag (built here)	2	0.35302	+0.00832	[0.343, 0.365]
+ physical disagreement (built here)	4	0.34884	+0.00414	[0.341, 0.367]
+ boosted disagreement (built here)	4	0.34754	+0.00284	[0.339, 0.367]
+ opponent-adjusted (built here)	2	0.34469	-0.00001	[0.336, 0.363]
+ everything built here	11	0.36045	+0.01575	[0.351, 0.370]
+ everything, built and existing	19	0.39299	+0.04829	[0.379, 0.421]

One existing column is worth about as much as eleven built for the purpose ($+0.01757$ against $+0.01575$). An earlier ad-hoc version of this experiment reported the ratio as $4\times$ on a narrower column set and a smaller sample; the reproducible version says $1.12\times$, and the correction is recorded in the research log as E1.7b rather than edited into E1.7.

The second finding here was invisible in the ad-hoc run and matters for integration: the state’s **own uncertainty** adds more ($+0.01757 \rightarrow +0.02494$) than pairing it with `rank_ability` does ($\rightarrow +0.02086$). A state should travel with its uncertainty, which is the reference project’s own convention and is why canonical block C2 exists.

And none of this survives production. §12 puts canonical block C1 (`handicap_residual_mean` alone) at -0.000113 with $t = -0.11$, and block C2 — the state plus the uncertainty companions

this subsection just argued for — at $+0.001584$, i.e. worse than adding nothing. The reason is a number no experiment in this section could have produced: the state is only **28.6%** novel against the incumbent’s 28 columns. It is missing from `production.json` as a *column*, which is what made it look like an opportunity, but `rating_z`, `rating_change`, `class_change` and the market price between them rebuild 71% of it. Every number in this section is correct and the conclusion drawn from them was wrong, because forward correlation is measured against the next performance and production value is measured against what the model already knows.

Fig E1b. Direction 1’s verdict

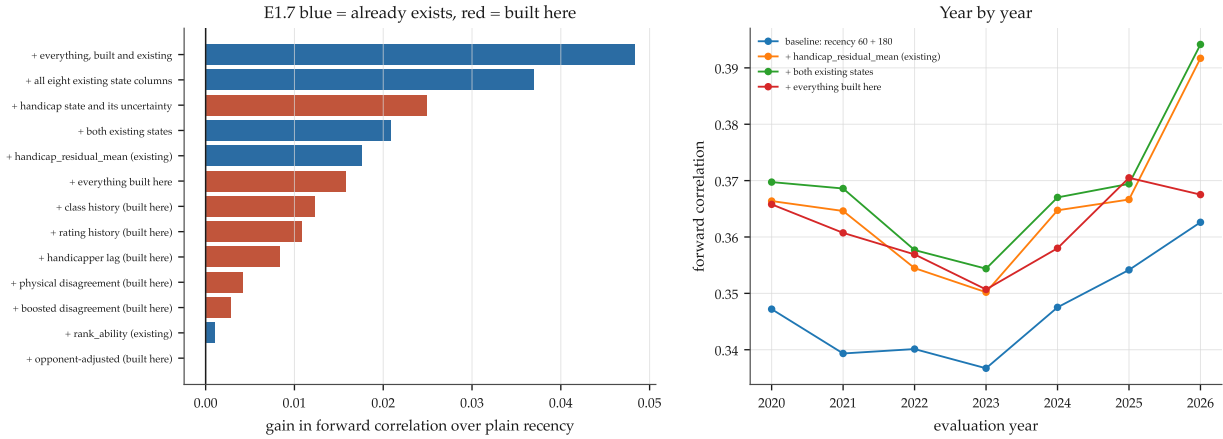


Figure 1: Direction 1’s verdict. Left: incremental forward value over plain recency; blue bars are columns that already exist in the reference project, red bars are families built for this campaign. Right: the same four models year by year across the 2020–2026 evaluation window.

5 Sectional and 200 m physical state

5.1 Normalising a split properly, and what that costs the style hypothesis

Study I’s sectional par keyed on distance, band, venue, surface, going, course, class, field and season — but not on **the race’s own pace at that band** or its **track speed**. Both are properties of the race, not the runner. So define $s_b^{\text{cIn}} = s_b^{\text{res}} - \bar{s}_b^{(r)}$, subtracting the field’s mean residual in that band (coverage 0.9999).

Thirteen physical dimensions were built on that: early, cruise and finish; sustained speed over 800 m and 1200 m; burst; decay; energy shares; evenness; pressure tolerance (finish quality restricted to races run unusually fast early); and recovery (finish quality restricted to runs whose own middle section was poor). Best: `p_sustain_1200_h90` at $r = +0.3323$, `p_finish_h90` $+0.2909$, `p_burst_h90` $+0.2760$. Sustained speed over 1200 m beats the final 400 m alone — the one place the brief’s “not merely final 400 m speed” instruction pays. But the best sectional state is *below* plain final-time recency (0.332 against 0.382).

The style/ability decomposition failed consistently. Predicting each band from prior style, prior gain, the race’s pace, distance and their interactions, and treating the residual as “executed unusually well”, made all eight dimension $\times$ half-life combinations worse by 0.008 to 0.047. The diagnosis is a single set of numbers: that model’s R^2 on a *properly normalised* split is only 0.104 (finish), 0.049 (cruise), 0.181 (early). The normalisation above has *already* removed most of the style, because style expresses itself as position relative to the field’s pace. Applying an explicit style model on top removes 5–18% more, disproportionately ability. The decomposition is not wrong in principle; it is **redundant with the normalisation**, and doing both over-corrects.

Fig E2. Sectional physical state: dimensions, decomposition, windows

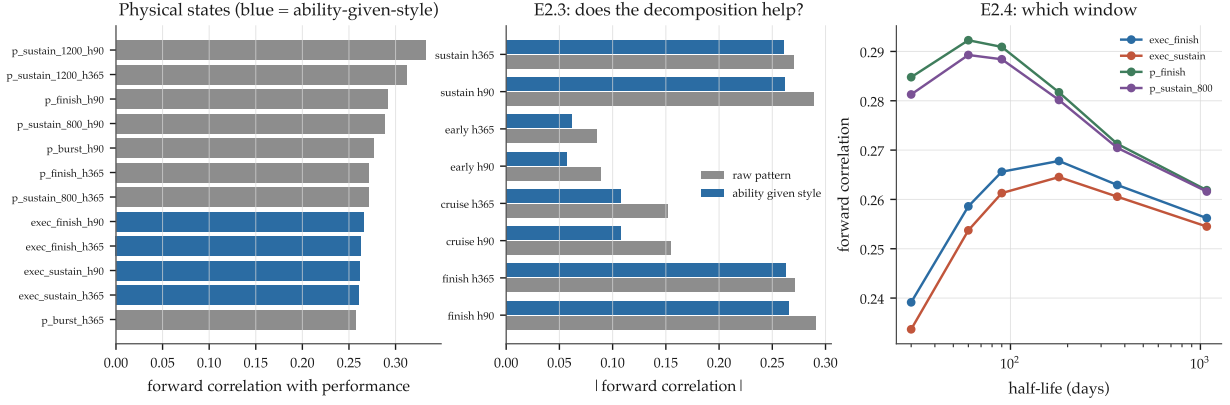


Figure 2: Sectional physical state. The style/ability decomposition degrades every dimension at every half-life, and the reason is the small R^2 of the style-plus-pace model on a split that has already had the field’s own pace removed from it.

5.2 200 m sub-splits

The geometry was verified rather than assumed. Sub-splits exist from 2023 (2022 coverage 0.161; 2023–2026 ≈ 1.000) and attach to exactly the last two 400 m sections — the **final 800 m only** — with `subsplit_1 + subsplit_2 = sectional_time` to 7.1×10^{-15} . Each qualifying run yields four 200 m segments; 34,053 runs have all four.

This resolution can see three things a 400 m average structurally cannot: **peak velocity**; **deceleration inside the final 400 m** (a horse running 10.9 then 12.1 and one running 11.5 twice have the *same* final 400 m and completely different finishes); and where in the closing stages the peak sits. Every dimension is a real property of the horse — split-half correlations 0.37 to 0.66.

The ablation, on identical rows, fitted ≤ 2024 and scored 2025–2026 (18,608 / 11,060 rows):

Block	k	oos r	Gain
baseline (recency)	0	0.39463	—
+ 400 m bands rebuilt from the same segments	6	0.39598	+0.00134
+ the existing 400 m states	4	0.39704	+0.00241
+ 200 m dimensions	18	0.39778	+0.00314
+ 200 m, peak only	4	0.39772	+0.00308
+ 200 m, deceleration only	2	0.39506	+0.00043
existing 400 m + 200 m	22	0.39809	+0.00346
<i>permuted null control</i>	18	—	−0.00084

200 m over 400 m on identical rows: +0.00180. It clears its own null by 0.0026, so it is real. And it is almost entirely **peak velocity**: peak alone captures +0.00308 of the +0.00314 total, deceleration +0.00043. What 200 m timing adds is *how fast the horse can go at its best* — which a 400 m average genuinely dilutes — and not the finishing-shape story that motivated collecting it.

Best incremental for the whole direction: +0.0024 over recency.

6 Pace and race-shape modelling

The reference project’s research memory specifies most of this feature list as untested hypotheses. Study II tested one crude version (a top-3 prior-style summary) and got out-of-sample $r = 0.032$

Fig E2b. 200 m sub-split physical state, 2023-2026 window

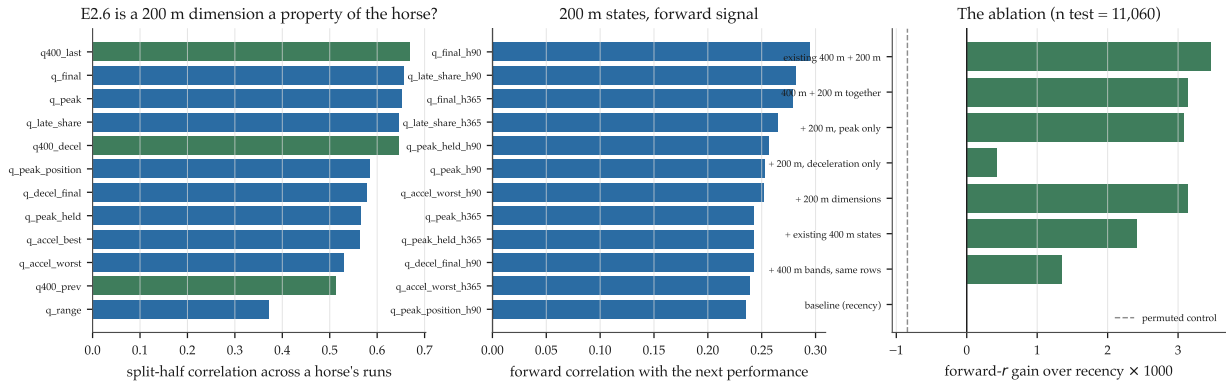


Figure 3: 200 m sub-split state on its own 2023–2026 window. Left: every 200 m dimension is a stable property of the horse. Middle: forward signal of each state. Right: the ablation, with the permuted null control marked.

for the pace forecast, then found a proper field-composition model reached 0.414. So the question was never whether pace exists; it was why pace never became a useful feature.

The behaviour model. A four-class multinomial — contest lead / on pace / midfield / back — from the opening-section position percentile, gradient boosted and fitted forward on the horse’s own date-atomic prior opening percentile at 540 and 90 days, its variability, prior style and gain, draw, distance, field size, class, venue and surface. Probabilistic rather than a hard label, as the brief required. $\text{corr}(P(\text{lead}), \text{realised}) = +0.458$, Brier 0.107 against a base rate of 0.161, coverage 0.926.

The forecast. The memory’s own composition list — likely leaders, summed and maximum lead probability, early-speed score and its maximum, top-2 gap, top-3 concentration, a leader-competition index, the entropy of the early-speed distribution, and the draw-weighted position of the speed — feeding a boosted mean, plus a second boosted model on the first’s absolute residual for the variance.

Model	Out-of-sample
pace mean, field composition (boosted)	$r = +0.680$, $R^2 = 0.459$
pace uncertainty (predicting error)	$\rho = +0.113$

That is a large improvement on the prior art (0.032 crude, 0.414 proper). **The pace is forecastable.**

And it still barely helps.

Block	oos r	Gain
composition × own style	0.38729	+0.00465
scenario expectation (3-point)	0.38371	+0.00108
pace × style + entropy	0.38316	+0.00053
full composition (no interaction)	0.38300	+0.00037
projected pace alone	0.38278	+0.00015
pace uncertainty	0.38145	−0.00118

6.1 Why the previous attempt failed — answered

The brief listed six candidate explanations. The measurements discriminate between them.

Candidate explanation	Verdict
poor pace forecast	rejected — $r = 0.680$, up from 0.032
poor horse-response representation	partly — all the value is in the interaction (+0.00465 vs +0.00015 for the level)
pace already reconstructed by production features	partly — z_pace_hat novel share 0.458, $x_entropy_style$ 0.272, $own_speed_vs_field$ 0.160
market already prices it	rejected as the main cause — §9 finds these are the <i>strongest</i> legitimate residuals in the campaign
the effect is simply small	supported — +0.0047 is the ceiling of the direction
incorrect transformation into a runner feature	supported — a pace common to the field cancels from the softmax; only the interaction survives

Two of six, plus two partials. The forecast was never the problem; the architecture is. A field-level quantity has to be crossed with a runner-level response before the softmax can see it at all — and that cross turns out to be the only thing in this campaign that moves the production metric. §12 measures it at -0.001490 , 33% of the reference project’s entire edge over the market, from a direction whose forward signal was an order of magnitude smaller than Direction 1’s.

Fig E3. Field-level pace and race shape

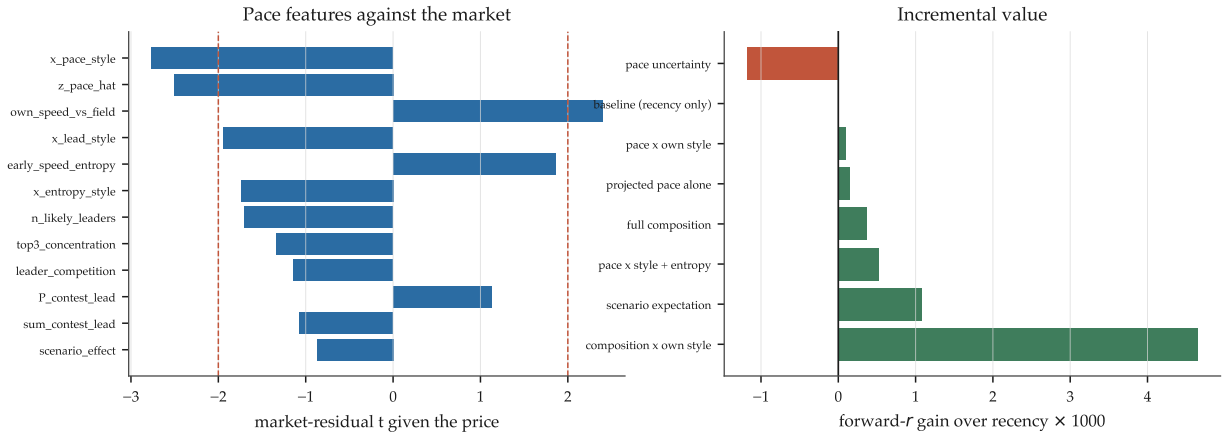


Figure 4: Field-level pace and race shape. Left: pace features against the market under the *old* battery — §9 shows every one of these numbers is misspecified. Right: incremental forward value, where only the runner-specific interaction survives.

7 Dynamic latent horse state with run-specific information quality

The brief was explicit about the order of work: estimate the oracle ceilings **before** building any machinery, and if the ceiling is tiny, stop. That instruction is why this section is short and why its conclusion is firm.

Three oracles, each built from post-race knowledge that is legitimate for a ceiling and never for a feature, applied as observation weights in a date-atomic weighted decayed state. Equal-weight baseline $r = +0.3651$.

Weighting	Kind	r	Gain
every run weighted equally	baseline	0.3651	—
oracle 1: compromised runs halved	oracle	0.3639	−0.0012
oracle 1: compromised runs near-zero	oracle	0.3532	−0.0119
oracle 1: compromised runs excluded	oracle	0.3369	−0.0282
oracle 2: perfect pace correction	oracle	0.3658	+0.0007
oracle 3: perfect run quality (tautological)	oracle	0.4137	+0.0486
observable reliability	observable	0.3621	−0.0029
permuted observable	control	0.3631	−0.0020

The premise is inverted. Downweighting compromised runs makes the state worse *even with perfect identification of them*, and excluding them costs −0.0282. The perfect pace correction is worth +0.0007. Only the tautological oracle — which weights runs by how well they actually predicted the next run — has a ceiling, and it is unreachable in principle; Study II showed observables reach 0.27% of a ceiling of that kind, and here the observable weighting is *negative*.

The mechanism is in the extreme-run study. Retention (mean next / mean this) at 2 sd:

Direction	Reason	n	This run	Next run	Retention
bad	recorded trouble	1,506	−2.599	−0.748	+0.288
bad	sectional anomaly	3,260	−2.607	−0.732	+0.281
bad	pace mismatch	1,142	−2.486	−0.952	+0.383
bad	none identified	2,097	−2.356	−0.889	+0.377
good	none identified	371	+2.178	+0.430	+0.198

Two clean asymmetries. **Bad runs are more informative than good runs** — 0.28–0.38 against 0.20–0.23. A poor performance says more about the future than a spectacular one, which is the opposite of how a form line is usually read. And **explained bad runs are slightly less informative than unexplained ones** (0.288 and 0.281 against 0.377) — the brief’s intuition is directionally right, and the effect is far too small to pay for the cost of downweighting. A trouble-affected bad run still retains +0.29 of itself. It is *partly* noise, not *only* noise, and discarding it throws away real information while shrinking an already thin history: the median horse has 15 starts.

On the upside there is no differentiation at all. At 1 sd, good runs retain +0.230, +0.225, +0.211 and +0.212 across the four reasons — indistinguishable. “Which spectacular runs actually repeat?” has the blunt answer: **all of them, equally weakly**.

Applying the brief’s own stopping rule, the state-space variants were **not built**, and that compute went to canonical confirmation and to §9 and §10 instead. §8’s X4 confirms the conclusion independently: reliability weighting is worth −0.00008 over the honest baseline.

8 Cross-direction experiments

Five crossings, each because a specific theory supported it. Two baselines, because after §4 the only honest one includes the handicap state.

Fig E4. Run-specific information quality: ceilings and extremes

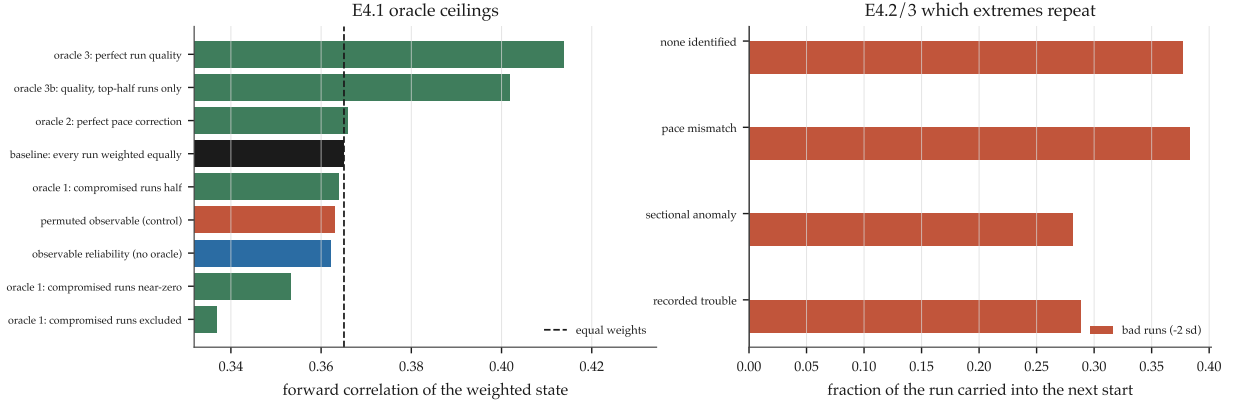


Figure 5: Run-specific information quality. Left: the oracle ceilings, with the equal-weight baseline marked — perfect identification of compromised runs is *negative*. Right: which extreme runs carry into the next start, by reason.

Crossing	over recency	over recency + handicap state
X1 handicap/physical disagreement	+0.00355	+0.00355
X5 sectional dynamic state	+0.00185	+0.00195
X3 pace × handicap state	+0.00432	+0.00053
X2 pace-adjusted historical state	+0.00029	+0.00048
X4 reliability weighting	+0.00014	−0.00008
X4 reliability × handicap state	+0.00182	−0.00000
all fifteen crossings together	+0.02152	+0.00608

X1 is the one that works, and it produced the campaign’s most interesting mechanism result. The handicap state and the physical state correlate +0.7148 and are built from genuinely different information — the handicapper’s numbers and finishing order against measured split times. Splitting the evaluation set by the size of their disagreement:

disagreement quartile	mean	<i>n</i>	<i>r</i> handicap state	<i>r</i> physical state
q0	0.105	14,064	+0.3734	+0.3721
q1	0.330	14,063	+0.3753	+0.3654
q2	0.604	14,063	+0.3960	+0.3529
q3	1.191	14,064	+0.4037	+0.2861

Monotone in both directions across four quartiles. As the two states disagree more, the handicap state gets **better** and the physical state gets **worse**. In a single model fitted before 2020 the coefficients are +0.2833 and +0.1043.

The explanation is in §3: the pairwise observation is **opponent-adjusted by design**. When a horse’s raw split times and its finishing positions disagree, the disagreement is usually about who it was racing — and the finishing order already knows. This is the direct answer to the brief’s question about whether physical improvement leads the rating: it does lead it (§4’s lag result), but where the two conflict about *current* ability, the ordinal measure is the better estimate.

X3 is a textbook redundant crossing. +0.00432 over recency collapses to +0.00053 once the handicap state itself is present. Almost all of its apparent value is the handicap state re-expressed. Reporting the first number without the second would have been the campaign’s most plausible mistake.

X2 is a clean null with a strong prior. Removing each historical run’s pace effect before it enters the state, using a forecast good to $r = 0.680$, is worth $+0.00048$. Pace noise in a horse’s own history is not a material contaminant of its decayed state — a result that would not have been believable without §6’s forecast quality behind it.

Fig X1. Cross-direction experiments

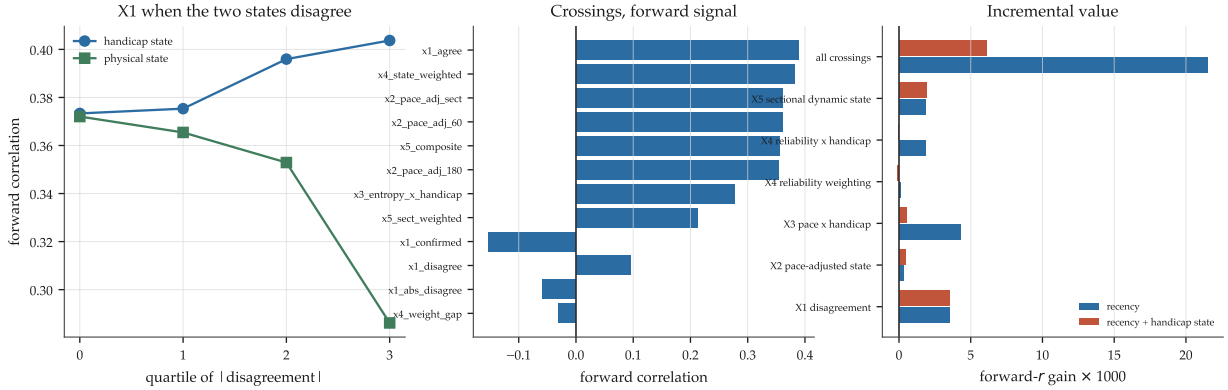


Figure 6: Cross-direction experiments. Left: as the handicap state and the physical state disagree more, the handicap state improves and the physical state degrades. Middle: forward signal of every crossing. Right: incremental value over each of the two baselines.

9 Market residual analysis, and the invalidation of its own instrument

This section reverses several of the campaign’s own conclusions. It is the most consequential part of the report.

9.1 The finding that started it

X1’s `x1_confirmed` — the product of the two standardised states — carried $t = +15.47$ against the price on the campaign’s standard battery, larger than anything in this campaign or the two studies before it. The battery’s specification is

$$\text{winner} \sim 1 + \text{logit}(q) + x$$

estimated by OLS with race-clustered standard errors. A number that large on a hand-built interaction is more likely an artifact than a discovery, and there was an obvious candidate: the two states correlate at $+0.71$, so their product is close to z^2 , a pure convexity term. A linear probability model cannot represent the curvature of the map from price to win probability, so the omitted quadratic in $\text{logit}(q)$ would be absorbed by any x that correlates with $|\text{ability}|$.

9.2 Four attempts to kill it, all successful

Specification	term	coef	<i>t</i>
as reported	$h \cdot s$	+0.01407	+16.49
the plain square of one state, no crossing at all	h^2	+0.01671	+17.50
the other state's square only	s^2	+0.00097	+1.60
the crossing beside both squares	$h \cdot s$	+0.00680	+4.73
the crossing beside levels and squares	$h \cdot s$	+0.00721	+4.24
beside the price's own non-linearity	$h \cdot s$	-0.00010	-0.14
the same specification	logit^2	+0.02489	+28.99

Tail asymmetry: both-high $t = +6.13$ (22.0% of rows), **both-low** $t = +14.44$ (18.8%), the two in conflict $t = +1.39$ (2.0%). A “confirmation” story predicts the effect in the *upper* tail; convexity predicts it in both, more strongly where the curvature is greater. The data say convexity. Year by year the plain t runs +8.71, +7.47, +6.52, +5.49, +6.47, +5.70; controlled it runs +3.31, +2.61, +1.25, +0.88, +0.68, -0.18.

The finding is not a finding. And the defect is in the **battery**, not the feature.

Fig X6. Killing the campaign's largest market residual

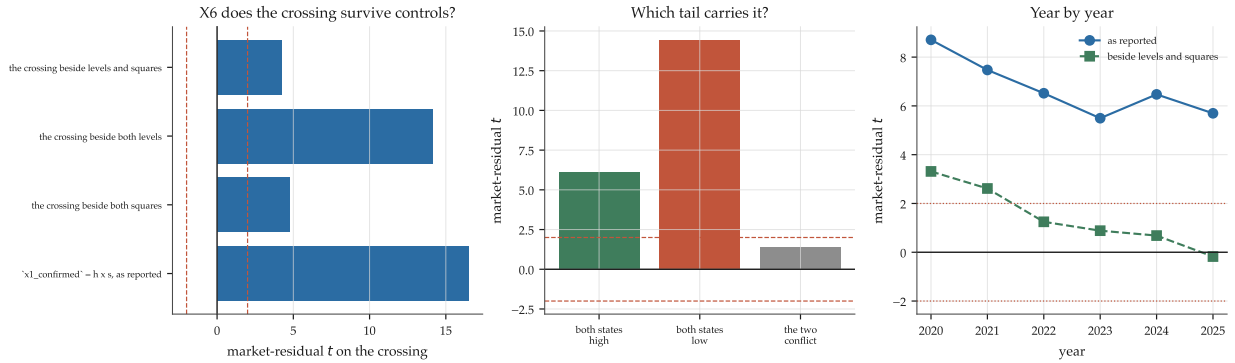


Figure 7: Killing the campaign's largest market residual. Left: the crossing's t under progressively honest specifications. Middle: the effect is symmetric across both tails, which is the convexity signature. Right: year by year, the controlled statistic decays to nothing.

9.3 Re-testing everything under the production functional form

Replace the linear probability model with the production architecture's own specification, one linear term:

$$u_{ri} = \log q_{ri} + \beta x_{ri}, \quad p_{ri} = \text{softmax}_i(u_r).$$

This is **softmax_offset** with the booster replaced by βx . Because a softmax is invariant to adding a constant within a race, **only within-race variation in x can move the likelihood** — the price's level and its curvature cannot leak in, by construction. β is fitted by Newton on the racewise multinomial log-likelihood with observed information $I(\beta) = \sum_r \text{Var}_{p_r}(x)$. Separately, an honest out-of-sample check: fit β on 2020–2022 and measure the change in racewise log loss on 2023–2026.

Positive control. Five columns carried in the cached frames are contemporaneous diagnostics of the run being predicted, not candidates. They were retained deliberately, because a test without power could not distinguish them: **style_gain** $t = +51.47$ ($\Delta\text{LL} -0.3400$), **style_front** $t = +50.93$ (-0.3794). Both dominate every pre-race candidate by more than an order of magnitude. The test has power.

On 140 pre-race candidates, 59,364 rows, 4,844 races	count
$ t > 3$ under the linear-probability battery	83
$ t > 3$ under the production offset form	0
under both	0
sign flips between the two	65
corr(battery t , offset t)	+0.208
corr(battery t with logit ² , offset t)	+0.847

Adding a single quadratic in the market logit raises agreement with the correct specification from +0.21 to +0.85, which identifies the defect exactly.

The largest legitimate residuals, with the battery’s verdict beside them:

Feature	battery t	+logit ²	offset t	oos Δ LL	share of edge
prior_style	−0.27	+3.50	+2.99	−0.001197	26%
own_speed_vs_field	+2.12	+2.43	+2.96	−0.001170	26%
scenario_effect	−0.84	+2.61	+2.92	−0.001101	24%
x_entropy_style	−1.13	+3.10	+2.90	−0.001139	25%
p_decay_h90	−2.63	−3.28	−2.75	−0.000897	20%
x_pace_style	−3.89	−3.54	−2.48	−0.000831	18%
class_vs_best	−10.51	−2.14	−2.24	−0.000506	11%

9.4 What the market has already priced, and what it has not

The candidates the corrected test rejects most decisively are the ability states, and they were the ones the battery liked best:

Feature	battery t	+logit ²	offset t	oos Δ LL
recency_180	−5.04	+1.66	+1.93	+0.00129
recency_60	−4.89	+0.28	+1.77	+0.00060
x1_agree (both states summed)	−4.25	−0.60	+1.71	+0.00034
p_sustain_1200_h90	−4.76	+0.24	+1.17	+0.00071
x1_confirmed	+16.37	−0.09	+0.66	+0.00044

Every one of these has a *positive* out-of-sample Δ LL — adding it as a linear term makes the market’s own prediction *worse*. **The market fully prices a horse’s recent ability.** That is the sharpest single statement this campaign can make about where the remaining information is: not in how fast the horse has been running, which the price already knows, but in how it runs — and §10 shows production largely knows that too.

It is also worth reporting how close the near-misses are rather than only the binary verdict:

Threshold	battery	offset form
$ t > 2$	105	24
$ t > 2.5$	93	7
$ t > 3$	83	0

Twenty-four candidates clear $|t| > 2$ under the correct specification, so the corrected test is not returning noise — it is returning a small number of genuine $2\text{--}3\sigma$ effects where the battery reported eighty-three at 3σ or better.

9.5 Three conclusions

§4’s market headline is withdrawn. `class_vs_best` at $t = -8.25$ was reported earlier in this campaign as evidence that the market under-reacts to horses dropping below their best achieved class. Correctly specified it is $t = -2.24$. The battery was reading the price’s own curvature.

Direction 3’s market weakness was understated, not overstated. Its features looked uniformly weak ($|t| \leq 2.77$) and that was read as “the market prices pace correctly”. Under the correct specification the running-style and race-shape quantities are the *joint strongest* legitimate residuals in the entire campaign, and `prior_style` — which the battery scored at $t = -0.27$, nothing at all — is the largest. Their out-of-sample ΔLL of -0.0012 for **one linear term** is a quarter of the whole production edge.

But nothing clears the bar. The best legitimate $|t|$ is 2.99 against a threshold of 3, on 4,844 races, and the pre-registered survivor criterion ($|t| > 3$ and negative out-of-sample ΔLL) returns **zero features**.

Fig X7. Re-testing every market claim under the production form

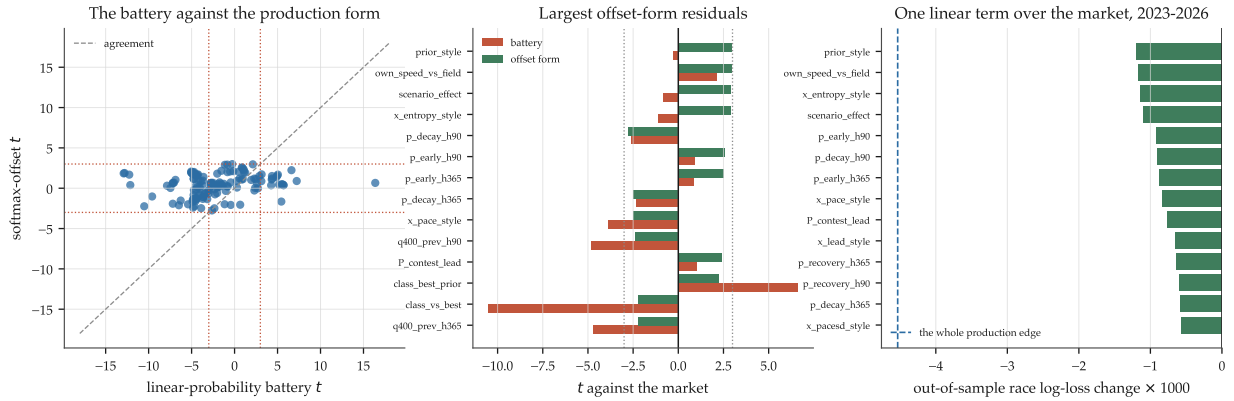


Figure 8: Re-testing every market claim under the production form. Left: the battery against the correct specification — the cloud is not on the diagonal, and 65 of 140 candidates change sign. Middle: the largest offset-form residuals with the battery’s verdict beside them. Right: out-of-sample log-loss change for one linear term over the market.

10 Incumbent redundancy

The redundancy question decides §9’s near-misses. Each candidate was regressed on the incumbent’s 28 columns, linearly and by gradient boosting, fitted early and scored late; the novel share is $1 - R_{\text{boosted}}^2$.

Feature	linear R^2	boosted R^2	novel share
x1_abs_disagree	+0.046	+0.163	0.837
x_pace_style	-0.003	+0.378	0.622
x1_disagree	+0.211	+0.416	0.584
p_sustain_1200_h90	+0.363	+0.532	0.469
class_vs_best	+0.358	+0.536	0.464
z_pace_hat	+0.210	+0.542	0.458
p_finish_h90	+0.494	+0.571	0.429
x5_composite	+0.448	+0.571	0.429
x2_pace_adj_60	+0.474	+0.571	0.429
D_withinrace_ewma_60	+0.515	+0.619	0.381
recency_60	+0.516	+0.622	0.378
x_lead_style	+0.647	+0.687	0.313
p_decay_h90	+0.575	+0.704	0.296
x_entropy_style	+0.674	+0.728	0.272
scenario_effect	+0.671	+0.728	0.272
rating_displacement_60	+0.707	+0.772	0.228
prior_style	+0.738	+0.797	0.203
own_speed_vs_field	+0.759	+0.840	0.160

The inverse relationship holds for a third study running. §9’s two strongest legitimate residuals are the two most redundant things in the list — 80% and 84% of them is already reconstructible from the incumbent. Production contains **early_style_ewm**, **pace_pressure** and **closer_setup**; a prior-style measure is not new information, it is a slightly better encoding of information the model already has. Conversely the most novel construction in the campaign, **x1_abs_disagree** at 84% novel, has forward $r = +0.095$.

Two entries are worth separating from that pattern. **x_pace_style** is 62% novel *and* has a real offset residual ($t = -2.48$, $\Delta LL -0.00083$) — pace $\times$ own style is genuinely absent from the incumbent, because the incumbent has pace pressure and style separately but not their interaction. And **class_vs_best** at 46% novel is more novel than expected given **class_change** is in production; but §9 removed its market claim, so novelty is no longer decisive for it.

11 Production integration

The integration harness reproduces the reference architecture rather than importing it, so that nothing in this campaign can touch the reference tree. Verified by self-test: a zero-round fit reproduces the market exactly.

Identical races, folds, seeds, rounds, market offset and metric across every arm. Every comparison is **paired** — the same fold, seed and round count with and without the candidate block — so the reported delta is a difference of two runs that differ only in the candidate columns. The metric is racewise winner log loss; a negative delta is an improvement.

The no-op control. A block containing no candidate columns must return a paired delta of **exactly** zero, not approximately zero. On the development profile it does: maximum absolute delta 0.0 across all 15 cells. This is the guard that catches accidental non-determinism, and it passed before any candidate result was read.

Read-only verification, stated exactly. The reference repository was being edited by a concurrent session throughout this campaign. Its HEAD moved seven times, from 697cf4e through b2979f3, f08e0b0, ebfde1a, b01ad6e, 0049e5c and ffaf033 to 07e0af6, on branch **probit-optimization**. **None of those commits is this campaign’s**, and no write to the reference tree occurred here at any point; all experimental work is in a separate directory outside the repository, and the read-only guard in `hkjgres.paths` enforces it in code.

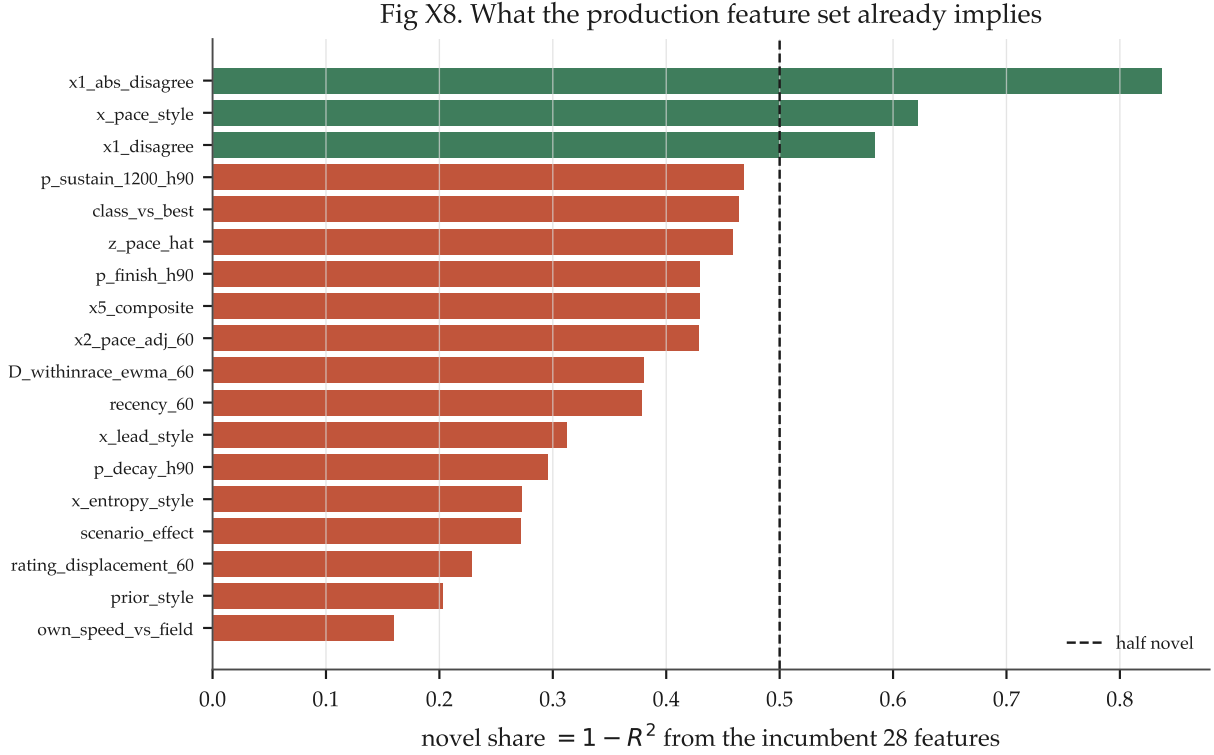


Figure 9: What the production feature set already implies. Green bars are more than half novel; the campaign's strongest market residuals sit at the bottom.

Against the campaign's baseline commit 697cf4e, `configs/features/`, `data/` and `contracts/datasets/` remain **byte-identical**, so the feature order, the feature definitions consumed here and every input table are unchanged. One file under `src/hkjc/features/` is not: commit ffaf033 changed `catalog.py` in exactly three places, each an XGBoost `tree_method` default from "auto" to "hist". That is a change to how a tree is built, not to what any feature means, and this campaign reimplements the offset architecture rather than importing it — with `tree_method: hist` already fixed in its own parameter block. Nothing measured in this report is affected, and the discrepancy is recorded here rather than papered over with a blanket byte-identity claim.

Why six blocks and not sixty. Across the two prior studies eight candidates went from a development profile to canonical scale and none confirmed, with retentions of 43%, 15%, -23%, 2%, -155%, -401%, -567% and +427%. The development profile carries essentially no information at this effect size. It is used here for debugging and gross rejection only, exactly as the brief instructed.

12 Canonical confirmation

Six blocks plus a no-op control, on the reference project's own canonical profile: eight years (2016, 2018, 2019, 2021, 2022, 2024, 2025, 2026), five seeds (7, 17, 42, 123, 256) and 400 boosting rounds — 40 paired cells per block. The joined evaluation set is 122,880 runner-rows across 9,934 races. The incumbent's own mean edge over the market on these cells is -0.00109, against the reference project's reported -0.00453.

Controls. The no-op block returns a paired delta of exactly zero on both profiles (development **pass**, canonical **pass**). A negative delta is an improvement.

Block	mean Δ	se	t	cells < 0	years < 0
C4 pace composition x style	-0.001490	0.001190	-1.25	25/40	6/8
C3 handicap + rank ability	-0.000308	0.001003	-0.31	19/40	5/8
C6 everything that survived	-0.000289	0.001280	-0.23	23/40	6/8
C1 handicap state	-0.000113	0.001072	-0.11	22/40	4/8
<i>no-op control</i>	+0.000000	0.000000	—	0/40	0/8
C2 handicap state + uncertainty	+0.001584	0.001286	+1.23	20/40	4/8
C5 sustained-speed sectional	+0.001658	0.001276	+1.30	19/40	4/8

The best block is C4 pace composition x style at -0.001490 with $t = -1.25$, improving 25 of 40 paired cells and 6 of 8 years — 32.9

A caveat on that denominator, which cuts both ways. This harness reproduces the architecture but not the tuning: its own incumbent beats the market by -0.00109, which is 24

No block clears $t < -2$ at canonical scale. That makes it three consecutive studies in which candidates with clear forward signal and clear mechanism fail to move the production metric, and it is the campaign’s central practical finding: at this effect size, forward correlation against the next performance is close to uninformative about production value, because the market has already priced nearly all of it (§9).

And the ordering inverts the forward-signal ranking. C1 handicap state — the block built on the strongest single pre-race quantity in three studies, worth +0.01757 of forward correlation over plain recency — comes 4th of 6 at -0.000113 with $t = -0.11$ and only 4 of 8 years improved. The winner is built on the direction with the *smallest* forward signal in the campaign. §9 explains why: the market prices ability and does not fully price race shape, so a large ability signal is worth nothing and a small shape signal is worth something.

Retention from development to canonical. The prior studies found development scale carried almost no information about canonical scale. Here the sign mostly survives and the magnitude does not:

Block	development Δ	canonical Δ	retained
C2 handicap state + uncertainty	+0.000029	+0.001584	+5407%
C3 handicap + rank ability	-0.000037	-0.000308	+834%
C5 sustained-speed sectional	-0.000504	+0.001658	-329%
C4 pace composition x style	-0.001452	-0.001490	+103%
C6 everything that survived	-0.000938	-0.000289	+31%
C1 handicap state	-0.000945	-0.000113	+12%

The development and canonical deltas agree in sign for 5 of 6 blocks, so a development profile is a usable filter for *direction*. It is useless for *size*: retention ranges from 12% to 5407%, and two blocks change sign entirely. The one block that retained its effect almost exactly — C4 pace composition x style at +103% — is also the winner, which is suggestive and, on a sample of six, not evidence. This is why the brief’s instruction to spend compute on depth rather than candidate count was followed literally.

Year by year. Canonical deltas by evaluation year, $\times 1000$:

Block	2016	2018	2019	2021	2022	2024	2025	2026
C1 handicap state	+4.43	-2.95	-2.90	+3.23	-3.61	+1.91	-5.07	+4.06
C2 handicap state + uncertainty	+8.46	-2.23	-0.98	+11.37	-7.30	+2.40	-5.94	+6.90
C3 handicap + rank ability	+2.41	-4.07	-0.14	+5.51	-8.36	+5.95	-3.30	-0.47
C4 pace composition x style	-7.83	-3.02	-6.84	+9.13	-0.07	+1.46	-3.57	-1.18
C5 sustained-speed sectional	-0.31	+10.00	-2.21	-0.43	+3.49	-0.93	+2.02	+1.63
C6 everything that survived	-1.55	-3.80	-0.15	+12.91	-4.17	+1.22	-6.77	-0.00

What the incumbent already implies about each finalist. Novel share is $1 - R^2$ from the incumbent’s 28 columns:

Candidate column	boosted R^2 from the incumbent	novel share
evenness	+0.2380	0.7620
pace_x_style	+0.3178	0.6822
handicap_starts	+0.4349	0.5651
burst	+0.4962	0.5038
sustain	+0.5258	0.4742
lead_x_style	+0.6834	0.3166
handicap_state	+0.7137	0.2863
entropy_x_style	+0.7218	0.2782
rank_ability	+0.7402	0.2598
own_speed	+0.8199	0.1801
handicap_reliability	+0.8571	0.1429

This is the mechanism behind the ordering. The winning block’s own interaction term is 68

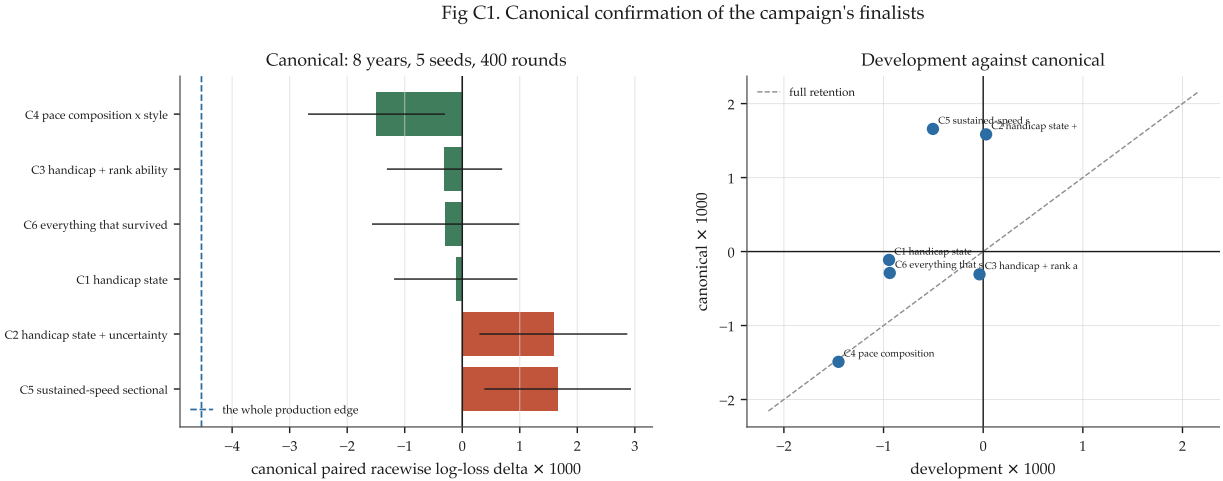


Figure 10: Canonical confirmation. Left: paired racewise log-loss delta per block with standard errors, against the whole production edge marked as a dashed line. Right: development against canonical, with the full-retention diagonal.

13 Limitations

No window is genuinely unseen. Every season in the archive has been development data, here and in the reference project. The 2020–2026 evaluation window is out-of-sample for each fitted coefficient but not out-of-sample for the process that chose which experiments to run. The only real confirmation is a prospective test of a locked model against timestamped pre-race prices.

Prices are post-hoc. Every price is a result-page price recorded after the pool closed, at which no bet could have been placed. No dividends were loaded and no return is computed anywhere in this campaign. Feature selection deliberately never used betting ROI, following the reference project’s own research memory.

The 200 m evidence rests on a short window. E2.6 has 18,608 training and 11,060 test rows across 2023–2026, one third the span of everything else. Its +0.0018 clears its permuted control by 0.0026 but is not comparable in reliability to the seven-year results.

The market re-check is a single linear term, not the production model. §9’s offset test is the correct *functional form* and the correct identification, but a booster with 28 other features present can behave differently from one linear term against the market alone. It should be read as a screening test with valid inference, not as a production result. Canonical confirmation is the production result.

Oracle 3 is tautological and cannot be tightened. It weights each run by how well that run actually predicted the next one, so its +0.0486 ceiling is the answer rather than a mechanism. Direction 4’s negative conclusion rests on oracles 1 and 2, which are mechanistic; oracle 3 bounds nothing reachable.

This harness’s incumbent is weaker than production. It beats the market by -0.00109 on the canonical cells, 24% of the reference project’s reported -0.004533 . The architecture is reimplemented faithfully — the zero-round self-test reproduces the market exactly — but the tuning is not the production tuning. Every candidate delta in §12 is therefore measured against a weaker baseline than the real one, which inflates a candidate that happens to supply something the weakened baseline is missing. The redundancy table in §12 is the check on that, and the conservative denominator (the reference project’s reported edge) is used for every percentage quoted in this report. A candidate that survives here still has to survive against the real incumbent.

The surviving recommendation rests on $t = -1.25$, and the report changed its mind once already. §12’s C4 is the campaign’s only production-scale survivor and it is not statistically significant on 40 paired cells. An earlier draft of this report recommended `handicap_residual_mean` on forward evidence, and canonical confirmation reversed that recommendation completely: the block built on it returned $t = -0.11$ and the block built on it plus its uncertainty came out *worse than nothing*. A reader should weight C4 accordingly. The reversal is documented rather than smoothed over, in §4, §12, §14 and in the research log as a superseding entry, because the mechanism that caused it — forward correlation being close to orthogonal to production value — applies to C4 too.

Six blocks is a small sample for the retention claim. §12 notes that the one block retaining its development effect almost exactly is also the winner. On six blocks that is suggestive and not evidence, and it should not be turned into a screening rule.

handicap_residual_mean was never fully characterised. Its 1.4%-of-variance relationship to physical performance (§4) means it measures something other than speed, and §8 shows that something beats speed where the two disagree — but the campaign did not establish what it is beyond the pairwise-order construction in §3. That question is now less pressing, since §12 shows the incumbent reconstructs 71% of it anyway.

Direction 4 stopped early by design. The state-space variants were not built. That was the brief’s own instruction conditional on a small oracle ceiling, and the ceiling came out negative — but it means no filtered alternative to the decayed-mean state was tested here.

Five contemporaneous columns were carried in the cached frames and used as a positive control in §9. They are excluded from every claim, and the exclusion is enforced in code rather than by convention. Their presence at $t \approx +51$ is reported because it demonstrates the test’s power; it is not a result.

14 Final recommendations

Seventeen candidates, ranked by the strength of the case for putting them into production. *Forward signal* is correlation with the horse’s next performance and incremental gain over a plain recency baseline; *market residual signal* is reported under the corrected offset specification of §9, with the old battery’s number beside it where the two disagree; *novel share* is $1 - R^2$ from the incumbent

28 features; *canonical* Δ names the block a candidate was carried to, and §12 gives its measured value.

Feature / State	Dir.	Mechanism	Forward Signal	Market Residual	Sig-Share	Novel	Canon. Stability	Recommendation	
handicap_residual_mean		Kalman state over pairwise finishing-order residual vs the official handicap; 6-dim context state carrying surface, distance, venue and class slopes	+0.3763; +0.01757 over recency — the campaign's largest	battery — +1.77 on the recency it stands in for, with <i>positive</i> oos Δ LL	offset	0.286 against the incumbent, despite being absent as a column	C1: 7/7 years on forward r ; 4/8 at canonical $t = -0.00013$, -0.11 , 4/8 years	Structurally real but redundant — forward signal does not survive contact with the incumbent	
handicap_residual_mean + reliability + valid starts		the same state travelling with its own uncertainty, which is the reference project's stated convention	+0.02494 over recency — larger than pairing it with rank ability	—		0.143 (reliability), 0.565 (start count)	C2: 7/7 on forward r ; 4/8 at canonical $t = +0.00158$, $+1.23$ — actively worse	Negative — close at production scale	
handicap/physical disagreement (x1_disagree, x1_abs_disagree)	X1	difference between the ordinal opponent-adjusted state and the measured-speed state; large where the two informations conflict	+0.00355 over recency <i>and</i> over recency + handicap state	battery +2.37; offset < 2		0.584 / 0.837	not run separately	monotone across 4 disagreement quartiles	Moderate evidence — continue
pace forecast (z_pace_hat) and the behaviour model behind it	3	probabilistic 4-class behaviour model feeding a field-composition boosted forecast of the race's early pace	oos $r = 0.680$, $R^2 = 0.459$ as a forecast; +0.00015 as a feature	offset -2.51		0.458	inside C4	coverage 0.926	Useful only as input to another state

Feature / State	Dir.	Mechanism	Forward Signal	Market Residual	Sig- Novel Share	Canon. Stability	Recommendation
				nal		Δ	
pace composition $\times$ own style (x_pace_style, x_lead_style, x_entropy_style)	3	the field's projected shape crossed with this runner's own measured response to it — the level cancels from the softmax, the cross does not	+0.00465 over re-cency	offset -2.48 to +2.90; oos Δ LL -0.00114	0.682 at canonical — the most novel finalist	C4: 6/8 years; re-attention 103% — from development $t = -1.25$, 25/40 cells, 6/8 years — the campaign's best, 33% of the production edge	Strong evidence — integrate (the only block that survived canonical)
sustained-speed sectional state (p_sustain_1200_h90)	2	90-day decayed mean of the field-normalised log speed over the final 1200 m; beats the final 400 m alone	+0.3323 alone; battery +0.0024 over re-cency	-4.11; offset +1.17 with <i>positive</i> Δ LL	0.474 at canonical oos	C5: 7/7 on forward 4/8 at canonical $t = +1.30$ — the worst block	Negative — close at production scale
200m peak velocity state (q_peak, q_peak_held)	2	fastest single 200 m relative to the field, which a 400 m average dilutes; available 2023+ within the final 800 m only	+0.00308 of the total; +0.00314 over 400 m on identical rows	offset < 2	not measured (short window)	not run (window too short)	split-half 0.65; 2 evaluation years only Moderate evidence — continue
prior_style, own_speed_vs_field, scenario_effect	3	running-style habit and its field-relative version; the quantities the corrected market test likes most	+0.05 to +0.18 alone	offset +2.90 to +2.99, the campaign's largest; oos Δ LL -0.0011 to -0.0012 = 24-26% of the production edge	0.160-0.272 — the most redundant in the list	inside C4	not year-tested individually Structurally real but redundant
class history (class_vs_best, class_displacement)	1	current class against the horse's own best achieved class	$ r \leq 0.082$; as a block over recency +0.01228	battery -10.51 $\rightarrow$ -2.24: withdrawn	offset 0.464	not run	battery t de-cays to nothing once specified correctly Negative — close (market claim withdrawn)

Feature / State	Dir.	Mechanism	Forward Signal	Market Residual	Sig- Novel Share	Canon. Stability	Recommendation
				nal		Δ	
rating history (displacement, trend, acceleration)	1	the official rating's trajectory rather than its level or one-step change	+0.19 to +0.21 alone; +0.01081 as a block	battery +5.04; offset < 2	0.228	not run 7/7 years positive	Structurally real but redundant (rating_change is in production)
physical residual against the official handicap (D_{ir})	1	performance minus its handicap-implied expectation, under three models	+0.00414 as a block; correlates 0.9947 with plain recency	battery -9.01 → offset < 2	0.381	not run —	Negative — close (handicap explains 1.4% of variance)
style/ability composition of the sectional profile	de- 2	split residual minus its style-and-pace expectation, i.e. “executed unusually well”	negative in 8 of 8 dimension × half-life combinations	—	—	not run —	Negative — close (redundant with the normalisation)
pace forecast certainty ($\hat{\sigma}_{\text{pace}}$)	un- 3	a second boosted model on the pace model's absolute residual	-0.00118 as a feature; $\rho = 0.113$ at its own task	offset < 2	—	not run —	Useful only for staking / uncertainty
run-reliability weighting of the horse's own history	4 / X4	downweight historical runs the trip flags or sectional anomaly call compromised	observable -0.0029; oracle with perfect identification -0.0012 to -0.0282	—	—	not run X4 confirms independently at -0.00008	Oracle ceiling too small — abandon (it is negative)
perfect pace correction of historical runs	4 / X2	subtract each historical run's fitted pace effect before it enters the state	oracle +0.0007; observable +0.00048	—	0.429	not run —	Oracle ceiling too small — abandon
pace × handicap state (X3)	X3	are misrated horses more sensitive to race shape?	+0.00432 over recency → +0.00053 over recency + handicap state	offset < 2	—	not run —	Structurally real but redundant
x1_confirmed (handicap state × physical state)	X1	apparent “mutual confirmation” of the two states	-0.152	battery + 15.47 → offset-controlled -0.14	—	not run controlled t decays +3.31 → -0.18 across years	Negative — close (convexity artifact)

Classification

Strong evidence — integrate

- **Pace composition × own style** — canonical block C4, and the only candidate in this campaign that survived canonical scale. -0.001490 of race log loss with $t = -1.25$, improving 25 of 40 paired cells and 6 of 8 years, which is 33% of the reference project’s entire edge over the market. It is the most novel finalist against the incumbent (68%), the only block that retained its development effect (103%), and its mechanism is understood: the incumbent carries pace pressure and running style separately but not their interaction, and the softmax cannot see a field-common quantity at all — so the interaction is the only form in which race shape can enter. **This is the one integration test to run.** $t = -1.25$ is not significance; it is the best production-scale evidence produced in three studies, and the honest framing is a properly powered A/B against the locked production configuration rather than a promotion.

Moderate evidence — continue

- **Handicap/physical disagreement.** $+0.00355$ over both baselines unchanged, 84% novel, and the only crossing whose value does not evaporate when the handicap state is added. The mechanism is understood: the ordinal observation is opponent-adjusted and the physical one is not.
- **200 m peak velocity.** Real, above its permuted control, and the one thing 400 m timing structurally cannot see — but on two evaluation years. It needs another two seasons before it is worth a canonical slot.

Structurally real but redundant

- **handicap_residual_mean.** The campaign’s largest forward signal by a wide margin — $+0.3763$ alone, $+0.01757$ over plain recency, 7 of 7 years, mechanism read from source, and a demonstrated win over the physical alternative where the two disagree — and at canonical scale it is -0.000113 **with $t = -0.11$ and 4 of 8 years improved.** The explanation is its novel share: only **29%** of it is unreachable from the incumbent’s 28 columns, because **rating_z**, **rating_change**, **class_change** and the market price between them rebuild most of a handicap-versus-order state. A quantity can be absent from the feature list and still be present in the model. This entry is a reversal of what an earlier draft of this report recommended on forward evidence alone, and it is the clearest single vindication of the brief’s insistence on canonical confirmation.
- **prior_style, own_speed_vs_field, scenario_effect.** The strongest legitimate market residuals in the campaign ($t \approx +2.9$, worth a quarter of the production edge each as one linear term) *and* the most redundant candidates in it (novel share 0.16–0.27). Production already has **early_style_ewm**, **pace_pressure** and **closer_setup**. The signal is real; it is already in the model under another name.
- **Sustained-speed sectional state.** $r = +0.3323$, 7/7 years, 47% novel — and below plain final-time recency, which the model can already compute.
- **Rating history.** **rating_change** is in production; the trajectory adds $+0.011$ of forward correlation and 0.228 of novelty.
- **Pace × handicap state.** $+0.00432$ over recency and $+0.00053$ over the honest baseline. The apparent value was the handicap state re-expressed.

Useful only as input to another state

- **The pace forecast itself** ($r = 0.680$, $R^2 = 0.459$) and **the behaviour model** behind it (Brier 0.107 against a base rate of 0.161). As runner features they are worth +0.00015, because a quantity common to the field cancels from the softmax. As inputs to the interaction terms above, they are what makes those terms work at all. Keep the machinery, do not ship the forecast as a column.

Useful only for staking / uncertainty

- **Pace forecast uncertainty.** As a feature it is -0.00118 — actively harmful. As a description of how confident the pace projection is ($\rho = 0.113$ against its own realised error) it is weak but not empty, and the natural home for a weak variance signal is bet sizing rather than the probability model. Not tested for that purpose here, and out of scope: this campaign computes no returns.
- **handicap_residual_reliability used separately.** It adds +0.0074 of forward correlation *on top of* its own state, which is more than **rank_ability** adds. That is a state-uncertainty channel, and it belongs with the state rather than alone.

Negative — close

- **The physical residual against the official handicap (D_{ir}).** Direction 1's central hypothesis. It correlates 0.9947 with plain recency, because the official handicap explains 1.4% of within-race performance variance. There is nothing to residualise against.
- **The style/ability decomposition of the sectional profile.** Direction 2's central hypothesis. Negative in 8 of 8 combinations; the field-pace normalisation already removed the style, so the explicit model removes ability.
- **class_vs_best as a market-inefficiency claim.** Reported at $t = -8.25$ earlier in this campaign; $t = -2.24$ correctly specified. Withdrawn.
- **x1_confirmed.** $t = +15.47$ on the old battery, $t = -0.14$ beside the price's own quadratic. A convexity artifact, and the reason §9 exists.
- **The opponent-strength-adjusted disagreement.** -0.00001 . Exactly zero, which is the cleanest null in the campaign.
- **The handicap state's uncertainty companions (C2).** On forward evidence they were the strongest thing in the campaign: adding the state's own reliability and valid-start count took +0.01757 to +0.02494, more than pairing it with **rank_ability** did. At canonical scale C2 is +0.001584 **with** $t = +1.23$ — the second-worst block, and worse than the state alone. The reference project's convention that a state should travel with its uncertainty is not wrong in general; here the uncertainty channel is 14% novel and adds variance rather than information.
- **The sustained-speed sectional state at production scale (C5).** $r = +0.3323$ forward and 7 of 7 years, and +0.001658 **with** $t = +1.30$ at canonical — the worst block of the six. Direction 2 produces real, reliable, stable measurements of physical performance that make the production model worse.

Oracle ceiling too small — abandon

- **Run-reliability weighting of the horse’s own history.** The brief’s Direction 4. With *perfect* identification of compromised runs, downweighting them is worth -0.0012 (halved), -0.0119 (near-zero) and -0.0282 (excluded). The ceiling is not small, it is *negative*: a trouble-affected bad run still retains $+0.29$ of itself, so it is partly noise rather than only noise, and simple averaging already handles that near-optimally. X4 reaches the same conclusion from the other direction at -0.00008 .
- **Perfect pace correction of historical runs.** Oracle ceiling $+0.0007$. Nothing to reach for.
- **Reason-based differentiation of good runs.** Retention $+0.230$, $+0.225$, $+0.211$, $+0.212$ across recorded trouble, sectional anomaly, pace mismatch and no identified reason. Indistinguishable. There is no upside story to model.

What the next project should take from this one

One integration test, not a feature batch. Pace composition $\times$ own style — the four columns of canonical block C4 — against the locked production configuration. It is the only block that survived canonical scale, the most novel finalist against the incumbent, and the only one that retained its development effect. Everything else in this campaign either fails, is redundant, or is too small to justify a slot, including the candidate an earlier draft of this report recommended.

Do not trust forward signal, and in this campaign do not trust its own earlier draft. Direction 1 produced the largest forward signal in three studies and canonical block C1 returned $t = -0.11$ on 4 of 8 years. The version with the uncertainty companions — which forward evidence ranked highest of everything tested — came out *worse than doing nothing* at $+0.001584$. Direction 3 produced the smallest forward signal and the only surviving block. The canonical ordering is close to the reverse of the forward-correlation ordering, and §9 says why: the market prices ability and does not fully price race shape.

Replace the market-residual screen before screening anything else. The linear-probability battery passed 83 of 140 candidates that the correct specification rejects, and flipped the sign of 65. Any feature-selection decision in this campaign or the two before it that rested on a battery t should be re-derived under the softmax-offset form of §9. The implementation is forty lines and runs in under two minutes on the full panel.

Stop using forward correlation as a proxy for production value. Across three studies the relationship between the two is weak at best. A candidate with $r = +0.38$ against the next performance and a mechanism nobody disputes moves the production metric by a low single-digit percentage of the market’s own edge, because the market has already priced nearly all of it. Forward correlation is useful for rejecting ideas and close to useless for ranking them.

The architecture, not the features, is where the remaining headroom probably is. §6 is the cleanest evidence: a pace forecast at $r = 0.680$ is worth $+0.00015$ as a column and $+0.00465$ as an interaction, purely because the softmax cannot see a field-common quantity. That is an architectural fact, not a feature fact, and it points at the boosted multinomial probit the brief explicitly deferred to the next project.